



JKBOSE 2026

CLASS 11TH

SEQUENCES AND SERIES

MATHS

LECTURE-03

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TOPICS TO BE COVERED

- 1 **Properties of S.P.**
- 2 **Arithmetic mean (A.M) and Geometric Mean (G.M.)**
- 3 **Relationship between A.M. and G.M.**
- 4 **Related Problems**

QUESTION (JEE-MAINS 2026)

$$\frac{a_9}{2^8}$$

$$\sqrt{2}^{10} = (2^{\frac{1}{2}})^{10} = 2^5 = 32$$



Let $a_1, \frac{a_2}{2}, \frac{a_3}{2^2}, \dots, \frac{a_{10}}{2^9}$ be a G.P. of common ratio $\frac{1}{\sqrt{2}}$. If $a_1 + a_2 + a_3 + \dots + a_{10} = 62$, then a_1 is equal to :

1 $\times 2 - \sqrt{2}$

2 $\times 2(2 - \sqrt{2})$

3 $\times \sqrt{2} - 1$

4 $\checkmark 2(\sqrt{2} - 1)$

(Sol) Here $\frac{a_2}{2a_1} = \frac{a_3}{2a_2} = \dots = \frac{a_{10}}{2a_9} = \frac{1}{\sqrt{2}}$

$$\Rightarrow \frac{a_2}{a_1} = \frac{a_3}{a_2} = \frac{a_4}{a_3} = \dots = \frac{a_{10}}{a_9} = \frac{2 \times \sqrt{2}}{\sqrt{2} \times \sqrt{2}}$$

$$\Rightarrow \frac{a_2}{a_1} = \frac{a_3}{a_2} = \frac{a_4}{a_3} = \dots = \frac{a_{10}}{a_9} = \frac{2\sqrt{2}}{2} = \sqrt{2}$$

$\Rightarrow a_1, a_2, a_3, \dots, a_{10}$ is a G.P. with

Common ratio $\sqrt{2}$.

Also $a_1 + a_2 + a_3 + \dots + a_{10} = 62$

$$\Rightarrow \frac{a_1(\sqrt{2}^{10} - 1)}{\sqrt{2} - 1} = 62$$

$$\Rightarrow a_1 \times \frac{1}{\sqrt{2}} = \frac{62}{\sqrt{2} - 1}$$

$$\Rightarrow a_1 = 2(\sqrt{2} - 1)$$

QUESTION (JEE MAINS 2026)



Suppose a, b, c are in A.P. and $a^2, 2b^2, c^2$ are in G.P. If $a < b < c$ and $a + b + c = 1$, then $9(a^2 + b^2 + c^2)$ is equal to 9.

(Sol) let $a = b - d, b = b, c = b + d$.

Here $a < b < c$ & $a + b + c = 1$

$$\Rightarrow d > 0 \text{ \& } b - d + b + b + d = 1$$

$$\Rightarrow 3b = 1$$

$$\Rightarrow b = \frac{1}{3}$$

Since $a^2, 2b^2, c^2$ are in G.P.,

$$\text{So } \frac{2b^2}{a^2} = \frac{c^2}{2b^2}$$

$$\Rightarrow 4b^4 = a^2 c^2$$

$$\Rightarrow 4b^4 = (b-d)^2 (b+d)^2$$

$$\Rightarrow 4\left(\frac{1}{3}\right)^4 = ((b-d)(b+d))^2$$

$$\Rightarrow \frac{4}{81} = (b^2 - d^2)^2$$

$$\Rightarrow \left(\frac{1}{9} - d^2\right)^2 = \frac{4}{81}$$

$$\Rightarrow \frac{1}{9} - d^2 = \pm \frac{2}{9} = \frac{2}{9}, -\frac{2}{9}$$

$$\Rightarrow -d^2 = \frac{2}{9} - \frac{1}{9}, -\frac{2}{9} - \frac{1}{9}$$

$$\Rightarrow -d^2 = \frac{1}{9}, -\frac{3}{9}$$

$$\Rightarrow d^2 = -\frac{1}{9}, \frac{3}{9}$$

$$\Rightarrow d^2 = \frac{3}{9} = \frac{1}{3}$$

$$\Rightarrow d^2 = \frac{1}{3}$$

$$\Rightarrow d = \frac{1}{\sqrt{3}} \quad (d > 0)$$

$$\text{So, } a = b - d = \frac{1}{3} - \frac{1}{\sqrt{3}}, \quad b = \frac{1}{3}, \quad c = \frac{1}{3} + \frac{1}{\sqrt{3}}$$

$$\Rightarrow 9(a^2 + b^2 + c^2) = 9 \left[\left(\frac{1}{3} - \frac{1}{\sqrt{3}} \right)^2 + \left(\frac{1}{3} \right)^2 + \left(\frac{1}{3} + \frac{1}{\sqrt{3}} \right)^2 \right]$$

$$= 9 \left[\frac{1}{9} + \frac{1}{3} - \frac{2}{3\sqrt{3}} + \frac{1}{9} + \frac{1}{9} + \frac{1}{3} + \frac{2}{3\sqrt{3}} \right]$$

$$= 9 \left(\frac{3}{9} + \frac{2}{3} \right) = \frac{9 \times 3}{9} + \frac{9 \times 2}{3} = 3 + 6 = 9$$

$$9(a^2 + b^2 + c^2) = 9$$

Arithmetic Mean (A.M.) and Geometric Mean (G.M.)

Arithmetic Mean: let a & b be any two positive numbers, then their Arithmetic mean is denoted & defined as;

$$A = \frac{a+b}{2}$$

e.g: $a=2, b=6$

$$\Rightarrow A = \frac{2+6}{2} = 4$$

$\Rightarrow A=4$ is the Arithmetic Means of 2 & 6.

* 2, 4, 6 is an Arithmetic Progression

In section of n -AM's b/w a & b :

Suppose that $A_1, A_2, A_3, \dots, A_n$ are n -AM's b/w a & b . Then $a, A_1, A_2, A_3, \dots, A_n, b$ is a Arithmetic progression with $(n+2)$ term

Here, $A_1 = a + d$

$$A_2 = a + 2d$$

$$A_3 = a + 3d$$

$\vdots$

$$A_n = a + nd$$

$\xrightarrow{(n+2)\text{ term}}$ & $b = a + (n+1)d$

$$\rightarrow b - a = (n+1)d$$

$$\Rightarrow d = \frac{b-a}{n+1}$$

Therefore $A_1 = a + \frac{1}{n+1}(b-a)$

$$A_2 = a + \frac{2}{n+1}(b-a)$$

$$A_3 = a + \frac{3}{n+1}(b-a)$$

⋮

$$A_n = a + \frac{n}{n+1}(b-a)$$

These are the required n -AM's b/w a & b .

Example: Insert 4 numbers b/w 1 & 31

So that the resulting sequence is an A.P

OR Find the 4-AM's b/w 1 & 31?

(Sol) Here $A_1 = 1 + \left(\frac{31-1}{5}\right) = 1 + 6 = 7.$

$$A_2 = 1 + \frac{2}{5}(31-1) = 1 + \frac{2 \times 30}{5} = 13$$

$$A_3 = 1 + \frac{3}{5}(31-1) = 1 + \frac{3 \times 30}{5} = 19$$

$$A_4 = 1 + \frac{4}{5}(31-1) = 1 + \frac{4 \times 30}{5} = 25.$$

Therefore the required A.P is:

1, 7, 13, 19, 25, 31

Hence the result.

Sum of n -AM's b/w a & b :

$$\text{let } S_n = A_1 + A_2 + A_3 + \dots + A_n$$

$$\Rightarrow S_n = \left(a + \frac{1(b-a)}{n+1}\right) + \left(a + \frac{2(b-a)}{n+1}\right) + \dots + \left(a + \frac{n(b-a)}{n+1}\right)$$

$$\Rightarrow S_n = na + \frac{b-a}{n+1}(1+2+3+4+\dots+n)$$

$$\Rightarrow S_n = na + \left(\frac{b-a}{\cancel{n+1}}\right) \times \frac{\cancel{n(n+1)}}{2}$$

$$\Rightarrow S_n = na + \frac{n}{2}(b-a)$$

$$\Rightarrow S_n = \frac{2na + nb - na}{2}$$

$$\Rightarrow S_n = \frac{na + nb}{2}$$

$$\Rightarrow S_n = n\left(\frac{a+b}{2}\right)$$

$$\Rightarrow \boxed{S_n = nA} \text{ , where } A = \frac{a+b}{2} \text{ is the arithmetic mean b/w } a \text{ & } b.$$

Geometric Mean: let a & b be any two positive numbers, then their Geometric mean is denoted and defined as;

$$G = \sqrt{ab}$$

e.g: $a = 4$ & $b = 9$

$$\Rightarrow G = \sqrt{4 \times 9} = \sqrt{36} = 6$$

$\Rightarrow 6$ is the G.M. b/w 4 & 9.

* 4, 6, 9 is a Geometric Progression.

Insertion of n -GM's b/w a & b :



Suppose that $G_1, G_2, G_3, \dots, G_n$ are n -GM's b/w a & b . Therefore, the resulting sequence $a, G_1, G_2, G_3, \dots, G_n, b$ is a Geometric Progression with $(n+2)$ terms.

Here $G_1 = ar$

$$G_2 = ar^2$$

$$G_3 = ar^3$$

$\vdots$

$$G_n = ar^n$$

Also, $b = ar^{n+1}$

$\nwarrow$
 $(n+2)^{\text{th}}$ Term

$$\frac{b}{a} = r^{n+1}$$

$$\Rightarrow r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$$

$$\text{So, } G_1 = a \left(\frac{b}{a} \right)^{\frac{1}{n+1}}$$

$$G_2 = a \left(\frac{b}{a} \right)^{\frac{2}{n+1}}$$

$$G_3 = a \left(\frac{b}{a} \right)^{\frac{3}{n+1}}$$

...

$$G_n = a \left(\frac{b}{a} \right)^{\frac{n}{n+1}}$$

These are n -GM's b/w a & b .

Example: Insert 3-GM's b/w 1 & 256.

or Insert 3-numbers b/w 1 & 256 so that the resulting sequence is a G.P.

(Sol) Here $G_1 = 1 \left(\frac{256}{1} \right)^{\frac{1}{4}} = (4^4)^{\frac{1}{4}} = 4$.

$$G_2 = 1 \left(\frac{256}{1} \right)^{\frac{2}{4}} = (16^2)^{\frac{1}{2}} = 16$$

$$G_3 = 1 \left(\frac{256}{1} \right)^{\frac{3}{4}} = (4^4)^{\frac{3}{4}} = 64$$

Therefore the required G.P. is:

1, 4, 16, 64, 256

Hence the result

Product of n -G.M's b/w a & b :

let P_n be the product of n -G.M's b/w a & b .

$$\text{So, } P_n = G_1 \times G_2 \times G_3 \times G_4 \times \dots \times G_n$$

$$\Rightarrow P_n = a\left(\frac{b}{a}\right)^{\frac{1}{n+1}} \times a\left(\frac{b}{a}\right)^{\frac{2}{n+1}} \times a\left(\frac{b}{a}\right)^{\frac{3}{n+1}} \times \dots \times a\left(\frac{b}{a}\right)^{\frac{n}{n+1}}$$

$$\Rightarrow P_n = a\left(\frac{b}{a}\right)^{\frac{1}{n+1} + \frac{2}{n+1} + \frac{3}{n+1} + \dots + \frac{n}{n+1}}$$

$$\Rightarrow P_n = a\left(\frac{b}{a}\right)^{\frac{1}{n+1}(1+2+3+\dots+n)} = a\left(\frac{b}{a}\right)^{\frac{1}{n+1} \times \frac{n(n+1)}{2}}$$

$$\Rightarrow P_n = a^n \left(\frac{b}{a}\right)^{\frac{n}{2}}$$

$$\Rightarrow P_n = a^n \times \frac{b^{\frac{n}{2}}}{a^{\frac{n}{2}}}$$

$$\Rightarrow P_n = a^{\frac{n}{2}} \times b^{\frac{n}{2}} = (a^{\frac{1}{2}} \times b^{\frac{1}{2}})^n$$

$$\Rightarrow P_n = \left[(ab)^{\frac{1}{2}}\right]^n$$

$$\Rightarrow P_n = (\sqrt{ab})^n = G^n$$

$\Rightarrow P_n = G^n$, where $G = \sqrt{ab}$ is the G.M. b/w a & b .

QUESTION (BLACK BOOK PROBLEM)

n arithmetic means are inserted between 7 and 49 and their sum is found to be 364, then n is:

1 11 ~~X~~

2 12 ~~X~~

3 13 ✓

4 14 ~~X~~

(Sol) We know that

$$S_n = nA$$

$$\Rightarrow 364 = n \times \left(\frac{7+49}{2} \right)$$

$$\Rightarrow n \times 28 = 364$$

$$\Rightarrow n = \frac{\cancel{364}^{52} \cdot 13}{\cancel{28}^4} \Rightarrow n = 13$$

Thank
You